

```
-- =====  
  
-- MODULE: PROJECT MANAGEMENT LAYER  
  
-- Database Engine: PostgreSQL \ Enterprise Configuration  
  
-- File Identity: · \_project_management_tables.sql  
  
-- Path: · \_Vertical_ERP_Modules/·V_Project_Management_Module/  
  
-- =====
```

```
CREATE EXTENSION IF NOT EXISTS pgcrypto;
```

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-- =====  
  
-- ١. projects (تعریف، حاکمیت و چرخه عمر پروژه‌ها در سطح مستأجر)  
  
-- =====
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```
CREATE TABLE projects (  
    project_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
    tenant_id UUID NOT NULL,  
    project_code VARCHAR(٥٠) NOT NULL,  
    name VARCHAR(٢٠٠) NOT NULL,  
    description TEXT,  
    start_date DATE NOT NULL,  
    end_date DATE,  
    actual_end_date DATE,  
    status SMALLINT NOT NULL DEFAULT ١, -- ١: Planning, ٢: Active, ٣: On Hold, ٤: Completed, ٥:  
Cancelled  
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
    updated_at TIMESTAMPTZ,  
    deleted_at TIMESTAMPTZ,
```

```
row_version BIGINT NOT NULL DEFAULT ۱
);
```

```
CREATE UNIQUE INDEX uq_projects_tenant_code ON projects(tenant_id, project_code) WHERE
deleted_at IS NULL;
```

```
CREATE INDEX idx_projects_status_lookup ON projects(tenant_id, status) WHERE deleted_at IS
NULL;
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-- ۲. project_tasks (ذیل پروژه WBS تسک‌ها، فعالیت‌ها و ساختار شکست کار)
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```
CREATE TABLE project_tasks (
    task_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    tenant_id UUID NOT NULL,
    project_id UUID NOT NULL REFERENCES projects(project_id) ON DELETE RESTRICT,
    parent_task_id UUID REFERENCES project_tasks(task_id) ON DELETE RESTRICT, -- ساختار درختی
    فعالیت‌ها
    wbs_code VARCHAR(۵۰) NOT NULL, -- (مثلا ۱,۱,۲) کد ساختار شکست کار
    title VARCHAR(۲۰۰) NOT NULL,
    description TEXT,
    planned_start_date DATE NOT NULL,
    planned_end_date DATE NOT NULL,
    actual_start_date DATE,
    actual_end_date DATE,
    progress_percentage NUMERIC(۵,۲) NOT NULL DEFAULT ۰,۰۰, -- پیشرفت فیزیکی از ۰ تا ۱۰۰ --
    priority SMALLINT NOT NULL DEFAULT ۲, -- ۱: High, ۲: Medium, ۳: Low
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        status SMALLINT NOT NULL DEFAULT ۱, -- ۱: Todo, ۲: In Progress, ۳: Review, ۴: Done, ۵:
Blocked
        created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
        updated_at TIMESTAMPTZ,
        deleted_at TIMESTAMPTZ
    );

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CREATE UNIQUE INDEX uq_project_tasks_wbs ON project_tasks(project_id, wbs_code) WHERE
deleted_at IS NULL;

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CREATE INDEX idx_project_tasks_hierarchy ON project_tasks(parent_task_id) WHERE
deleted_at IS NULL;

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-- =====
-- ۳. project_members (اعضای پروژه و ماتریس مسئولیت‌ها و نقش‌های تیم کاری)
-- =====
-- Replace project_members table in ۰۱_project_management_tables.sql

```

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CREATE TABLE project_members (
    project_member_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    tenant_id UUID NOT NULL,
    project_id UUID NOT NULL REFERENCES projects(project_id) ON DELETE RESTRICT,
    employee_id UUID NOT NULL, -- Logical Reference to hr.employees
    project_role VARCHAR(۱۰۰) NOT NULL,
    joined_at DATE NOT NULL DEFAULT CURRENT_DATE,
    left_at DATE,
    is_active BOOLEAN NOT NULL DEFAULT TRUE,

```

```

-- Standard Audit & Soft Delete Fields
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
created_by UUID,
updated_at TIMESTAMPTZ,
updated_by UUID,
deleted_at TIMESTAMPTZ,
deleted_by UUID,
row_version BIGINT NOT NULL DEFAULT 1
);

```

```

-- Corrected Unique Index with Soft Delete Policy
CREATE UNIQUE INDEX uq_project_members_unique
ON project_members(project_id, employee_id)
WHERE deleted_at IS NULL;

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```

CREATE INDEX idx_project_members_tenant
ON project_members(tenant_id)
WHERE deleted_at IS NULL;

```

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-- ٤. resource_allocations (تخصیص منابع فیزیکی، ماشین آلات یا پرسنل انسانی به فعالیت‌ها)

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-- Replace resource_allocations table in ١\project_management_tables.sql

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CREATE TABLE resource_allocations (
allocation_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

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tenant_id UUID NOT NULL,
task_id UUID NOT NULL REFERENCES project_tasks(task_id) ON DELETE RESTRICT,
resource_type SMALLINT NOT NULL, -- 1: Human, 2: Equipment/Machine, 3: Material
resource_id UUID NOT NULL, -- Polymorphic Logical Reference
allocated_quantity NUMERIC(10,4) NOT NULL DEFAULT 1,0000, -- Standardized Precision
start_date DATE NOT NULL,
end_date DATE NOT NULL,

-- Audit & Soft Delete Fields
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
created_by UUID,
updated_at TIMESTAMPTZ,
updated_by UUID,
deleted_at TIMESTAMPTZ,
deleted_by UUID,
row_version BIGINT NOT NULL DEFAULT 1
);

```

```

CREATE INDEX idx_resource_allocations_task
ON resource_allocations(task_id)
WHERE deleted_at IS NULL;

```

```

CREATE INDEX idx_resource_allocations_tenant
ON resource_allocations(tenant_id)
WHERE deleted_at IS NULL;

```